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Pablo Miralles Ph.D. Candidate

Scholar GitHub LinkedIn

Research Interests

I work on adapting large language models to downstream tasks in low-resource settings in a way that generalizes. My work combines parameter-efficient reinforcement-learning fine-tuning (QLoRA + GRPO for natural language inference) with training-free methods built on LLM log-probabilities (AI-generated text detection, authorship attribution and verification), with a focus on robustness and out-of-distribution generalization.

Education

Technical University of Madrid — Madrid, Spain
Ph.D. in Artificial Intelligence — 2024 – Now
- Advisor: Prof. Alejandro Martín. Expected defence: September 2027.
- Thesis theme: adapting LLMs to low-resource downstream tasks in a way that generalizes (working title pending).

Technical University of Madrid — Madrid, Spain
M.Sc. in Machine Learning and Big Data — 2023 – 2024
- GPA: 9.65/10.00, Rank: 1st

University of Murcia — Murcia, Spain
B.Sc. in Mathematics — 2017 – 2022
- GPA: 9.74/10.00, Rank: 1st

University of Murcia — Murcia, Spain
B.Sc. in Computer Science — 2017 – 2022
- GPA: 9.57/10.00, Rank: 2nd

Research Visits

Queen Mary University of London — London, UK
Visiting Ph.D. Researcher — 2026.09 – 2026.12
- Host: Prof. Arkaitz Zubiaga. Research on multimodal hate speech detection.

Publications

Conference papers

1. **Miralles-González, P.**, Huertas-Tato, J., Martín, A., & Camacho, D. (2026). One-Shot Style Transfer LLM Log-Probabilities for Authorship Attribution and Verification. *Findings of the Association for Computational Linguistics: EMNLP 2026*, 278–297, Budapest, Hungary. https://doi.org/10.18653/2026.findings-emnlp.14 GitHub

Journal articles

1. **Miralles-González, P.**, Huertas-Tato, J., Martín, A., & Camacho, D. (2026). Evaluating the scalability and adversarial generalization of GRPO-trained NLI models. *Knowledge-Based Systems*, 352, 117013. https://doi.org/10.1016/j.knosys.2026.117013 GitHub
2. **Miralles-González, P.**, Huertas-Tato, J., Martín, A., & Camacho, D. (2026). Not all tokens are created equal: Perplexity Attention Weighted Networks for AI-generated text detection. *Information Fusion*, 125, 103465. https://doi.org/10.1016/j.inffus.2025.103465 GitHub
3. **Miralles-González, P.**, Huertas-Tato, J., Martín, A., & Camacho, D. (2026). On the locality bias and results in the Long Range Arena. *IEEE Transactions on Emerging Topics in Computational Intelligence* (accepted, to appear). https://doi.org/10.48550/arXiv.2501.14850 GitHub

Preprints

1. Girón, A., **Miralles-González, P.**, Huertas-Tato, J., D'Antonio, S., & Camacho, D. (2026). xList-Hate: A Checklist-Based Framework for Interpretable and Generalizable Hate Speech Detection. *arXiv* (under review). https://doi.org/10.48550/arXiv.2602.05874

Teaching

Generative Models | M.Sc. in Deep Learning (Instructor) 2024/25 – present
- VAEs, GANs, autoregressive and diffusion models. Co-designed syllabus and materials (50%) with another instructor; 15–30 students per edition.

Deep Learning for NLP | M.Sc. in Deep Learning (Instructor) 2024/25 – present
- Real-world NLP applications with HuggingFace, LangGraph, vLLM and vector databases. Co-designed syllabus and materials (50%) with another instructor; 15–30 students per edition.

Data Structures | B.Sc. in Computer Science (Teaching assistant) 2025/26 – present
- Queues, stacks, lists, trees, graphs and hash tables.

Classical Prediction Methods | B.Sc. in Data Science (Teaching assistant) 2025/26
- Lectures and practical sessions on linear regression, clustering and time series models.

Algorithms and Data Structures | B.Sc. in Data Science (Teaching assistant) 2024/25
- Graph theory fundamentals and greedy algorithm design.

Mentoring and Service

Supervision: Co-supervised 5–6 M.Sc. theses.
Seminars: Co-organizer of the weekly research group seminar series (2025/26): members' research talks and tutorials on tools and topics such as LLM efficiency.
Peer review: IDEAL (2 papers), The Computer Journal (1 paper).

Projects and grants

FPU Fellowship (University Teacher Training Programme)
Spanish Ministry of Science and Innovation 2025.01 – Now
- Highly competitive predoctoral contract aimed at training future university lecturers and researchers.

XAI-DisInfodemics: eXplainable AI for disinformation and conspiracy detection during infodemics
European Union NextGenerationEU/PRTR 2024.05 – 2025.01

FightDIS: Fighting against Information DISorders in Online Social Networks
Spanish Ministry of Science and Innovation 2023.09 – 2024.05

Awards and Honors

- **Best academic record**, M.Sc. in Machine Learning and Big Data 2024
- **Extraordinary End of Degree Award**, B.Sc. in Mathematics and B.Sc. in Computer Science 2023
- **Academic Excellence Award in joint B.Sc. programs**, Math + CS 2023
- **SWERC (ICPC regional)**: 18th, 17th and 23rd place, 2019/20–2021/22 2020–2022

Skills

Languages: Spanish (native), English (C1).
Programming: Python, C++, Java.
Deep Learning: PyTorch, HuggingFace, vLLM, QLoRA, GRPO, Triton (basic), CUDA (basic).
Data Science/Machine Learning: scikit-learn, NumPy, Pandas.
Experiment management/distributed: Weights & Biases, Ray.
Other: Docker, Linux, Git, LaTeX.

Industry Experience

Fever | Software Engineer 2023.02 – 2023.08
- Led a project using LLMs to automatically structure information and generate plan content, substantially reducing content production costs.